GLOBAL SECTIONS OF A DERIVED TENSOR PRODUCT WITH A CONSTANT COMPLEX OF ABELIAN GROUPS PULL THE CONSTANT FACTOR OUT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the torsion free representative justified through global dimension one, the truncation reductions using the two displayed vanishing statements, and the final reduction to a finite free module by Lazard's theorem and the colimit theorem.

Editorial note. The label lemma-pull-out-constant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constant sheaf underline K associated to a complex of abelian groups. Same chapter, the constant sheaf on X_etale associated to a complex K of abelian groups and its derived tensor product with an object of D(X_etale). Supplies the object E tensor^L_Z underline K whose global sections are computed.
- K-flat resolutions over Z and the last-one-flat criterion. More on Algebra, the existence of K-flat resolutions and the flatness criterion used to replace K by a bounded below complex of torsion free modules. Supplies the representative K^bullet = tau_{>= b-1}L^bullet of K by torsion free modules, using that Z has global dimension one.
- Lazard's theorem and the colimit theorem for etale cohomology. Algebra, the description of a flat module as a filtered colimit of finite free modules, and the same chapter's theorem computing cohomology of filtered colimits. Supplies the reduction from a single flat module to a finite free module in the last step of the proof.

Source reference. Original theorem 058G: sources/stacks-chow/tags.txt, source lines 6779–6779. Prerequisite pointer invoked at proof line(s) 18995; it names a

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standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0653: sources/stacks-chow/tags.txt, source lines 7887–7887. Prerequisite pointer invoked at proof line(s) 18955; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Prerequisite pointer invoked at proof line(s) 18996; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $E \in D^+(X_{étale})$ and $K \in D^+(\mathbf{Z})$. Then*

$$R\Gamma(X, E \otimes^{\mathbf{L}}_{\mathbf{Z}} \underline{K}) = R\Gamma(X, E) \otimes^{\mathbf{L}}_{\mathbf{Z}} K$$

Proof. Say $H^i(E) = 0$ for $i \geq a$ and $H^j(K) = 0$ for $j \geq b$. We may represent K by a bounded below complex $K^\bullet$ of torsion free $\mathbf{Z}$-modules. (Choose a K-flat complex $L^\bullet$ representing K and then take $K^\bullet = \tau_{\geq b-1} L^\bullet$. This works because $\mathbf{Z}$ has global dimension 1. See More on Algebra, Lemma 0653.) We may represent E by a bounded below complex $\mathcal{E}^\bullet$. Then $E \otimes^{\mathbf{L}}_{\mathbf{Z}} \underline{K}$ is represented by

$$\mathrm{Tot}(\mathcal{E}^\bullet \otimes_{\mathbf{Z}} \underline{K^\bullet})$$

Using distinguished triangles

$$\sigma_{\geq -b+n+1} K^\bullet \to K^\bullet \to \sigma_{\leq -b+n} K^\bullet$$

and the trivial vanishing

$$H^n(X, \mathrm{Tot}(\mathcal{E}^\bullet \otimes_{\mathbf{Z}} \sigma_{\geq -a+n+1} \underline{K^\bullet}) = 0$$

and

$$H^n(R\Gamma(X, E) \otimes^{\mathbf{L}}_{\mathbf{Z}} \sigma_{\geq -a+n+1} K^\bullet) = 0$$

we reduce to the case where $K^\bullet$ is a bounded complex of flat $\mathbf{Z}$-modules. Repeating the argument we reduce to the case where $K^\bullet$ is equal to a single flat $\mathbf{Z}$-module sitting in some degree. Next, using the stupid trunctions for $\mathcal{E}^\bullet$ we reduce in exactly the same manner to the case where $\mathcal{E}^\bullet$ is a single abelian sheaf sitting in some degree. Thus it suffices to show that

$$H^n(X, \mathcal{E} \otimes_{\mathbf{Z}} \underline{M}) = H^n(X, \mathcal{E}) \otimes_{\mathbf{Z}} M$$

when M is a flat $\mathbf{Z}$-module and $\mathcal{E}$ is an abelian sheaf on X. In this case we write M is a filtered colimit of finite free $\mathbf{Z}$-modules (Lazard's theorem, see Algebra, Theorem 058G). By Theorem 09YQ this reduces us to the case of finite free $\mathbf{Z}$-module M in which case the result is trivially true. □